

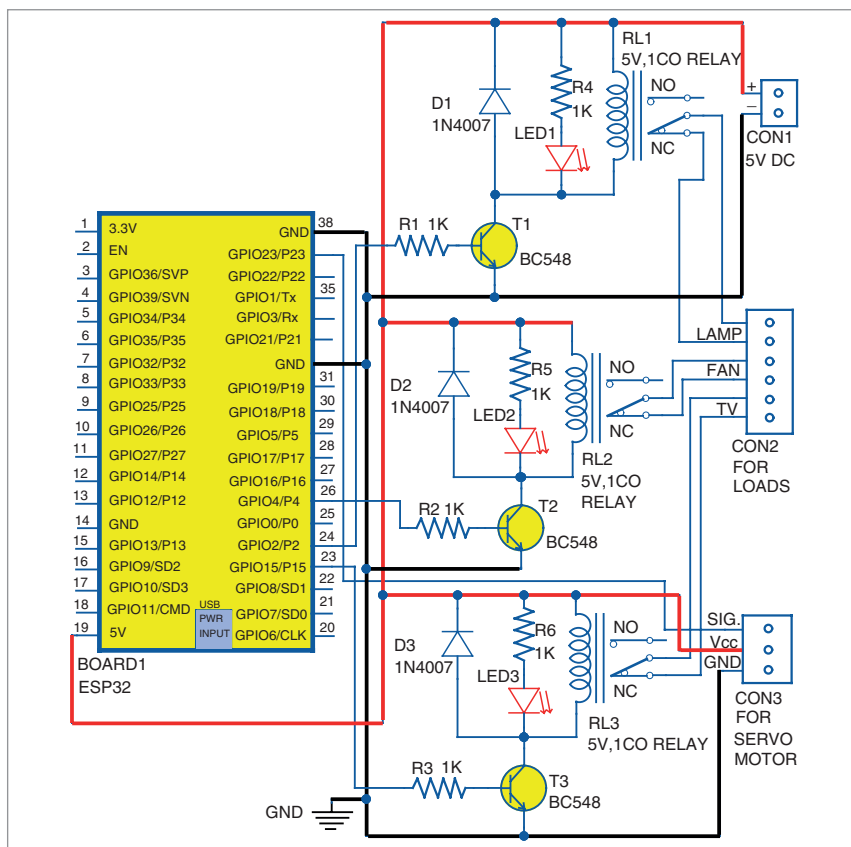


Fig. 4: Circuit diagram

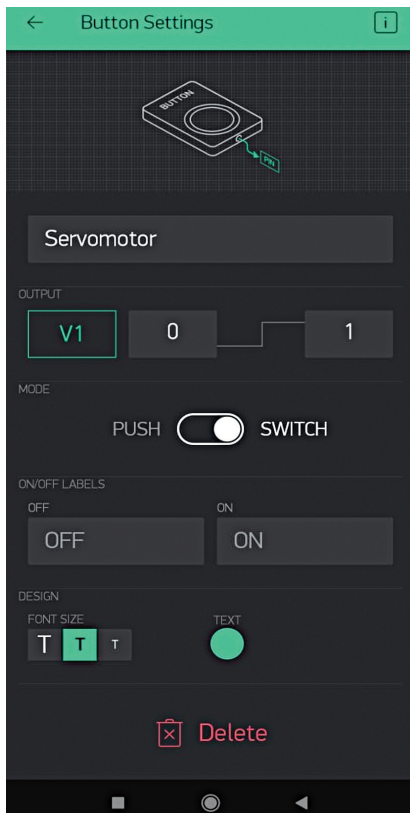


Fig. 5: Servo motor's setting in Blynk app

digital output as gp4 and gp15. Here, gp2 corresponds to lamp/light connected to GPIO2 of ESP32. Similarly, gp4 and gp15 are connected to GPIO4 and GPIO15 of ESP32 for fan and TV, respectively.

8. For servo motor, rename Button as Servomotor. Click the output box, select Virtual pin as V1 and change Mode to Push option (Fig. 5). Similarly, select Button again and rename it as ServoV2, output box as V2, and Mode as Switch option.

9. When you are done with the above settings, press the Play button. This will switch from Edit mode to Play mode where you can interact with the hardware. While in Play mode, you won't be able to drag or



Fig. 6: IFTTT on Android phone

set up new widgets. Press Stop and get back to Edit mode.

10. Next, open Arduino IDE, download the Blynk library from the link <https://github.com/blynkkk/blynk-library>

Download this zip file, extract it, and then copy this library file to Arduino library files. After installation, you can find your Blynk library file → examples → Blynk → Boards_WiFi → ESP32_WiFi. Insert the Auth Token key, your Wi-Fi details in the Arduino code (bedridden.ino), compile the Arduino code and upload it to ESP32 board.

For controlling the appliances over the Internet using Google Assistant you need to do the following:

1. Go to website of IFTTT (<https://ifttt.com>). Install it on your mobile phone.
2. Signup using your Google account (the same Google account that you are using with the Google Assistant).
3. Once you are signed in, click on 'My Applets' and select 'New Apple' which will trigger relay to turn on the light.

4. Now click on 'If This' as shown in Fig. 6 and, in the search bar, search for 'Google Assistant' and tap on it. A window will pop up as shown in Fig. 7.

5. Select 'Say a simple phrase' option for the triggering action. Click on Connect to give permission to trigger action.

6. Next, in the first text box, type the phrase that you would want Google Assistant to trigger the action. For example, "Turn on the light."

7. For the next two text boxes, you write alternate commands like "Switch on the light," and "Light on" for the same action.

8. In the fourth text box, type the response to be given by Google Assistant, for example, "Okay, I will turn on the light." Click on 'Create Trigger' button.

9. Next click on 'Then that' button.

10. Type 'webhooks' in the search box, select it and click Connect. Webhooks will allow us to send commands to the Blynk server. To do that you